

DES PUNE UNIVERSITY
School of Engineering and Technology
Computer Engineering and Technology
Program: B.Tech. Computer Science and Engineering

Academic Year: 2025-26	Year: Second Year	Term: I
Roll No.: 1012411011	Name: Riddhee Kulkarni	
Subject: Java		
Assignment No.: 8	Title: File Handling	
Date: 17/09/2025		

AIM:

To write a program to read from and write to a file using FileReader, FileWriter, BufferedReader, and BufferedWriter, applicable to any system that requires data storage and retrieval.

THEORY:

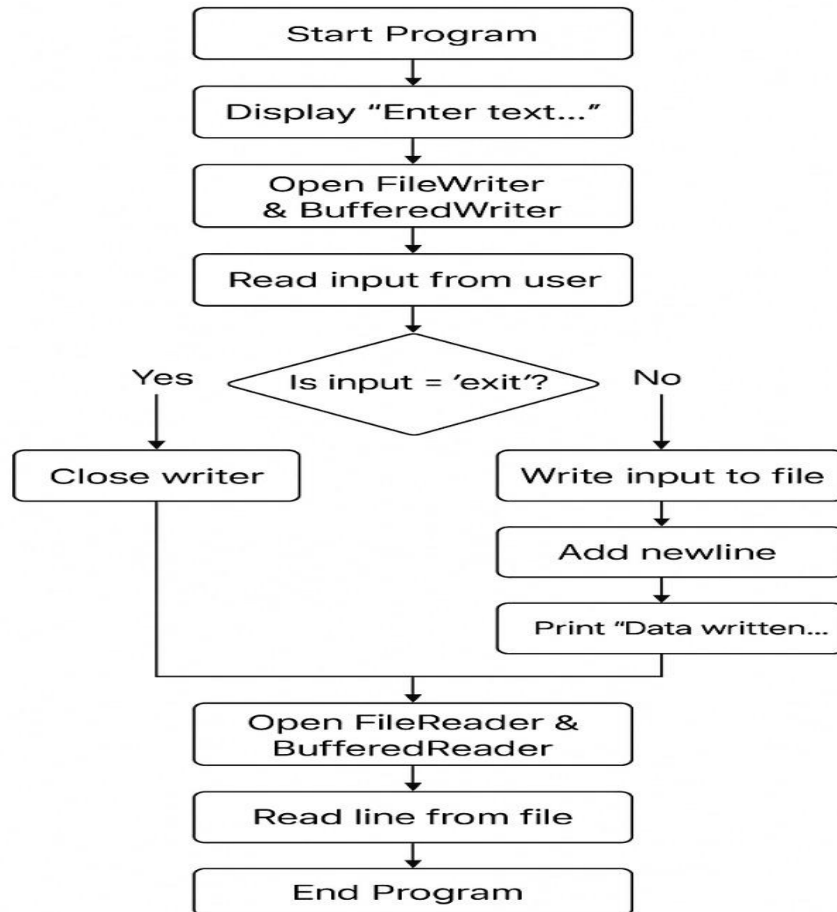
File handling in Java is a mechanism that allows programs to create, read, write, and manage files on disk. It is essential for storing data permanently so that it can be retrieved and reused later. File handling is achieved through the classes of the java.io package.

Streams are used to establish communication between the program and the file. An input stream is used when data flows from the file to the program, and an output stream is used when data flows from the program to the file.

FileWriter is used to write text data to a file, and BufferedWriter is used on top of it to improve efficiency by providing a buffer and methods like `newLine()` for line breaks. Similarly, FileReader is used to read text data from a file, while BufferedReader provides efficient reading using a buffer and supports reading a complete line at once with the `readLine()` method.

DES PUNE UNIVERSITY
School of Engineering and Technology
Computer Engineering and Technology
Program: B.Tech. Computer Science and Engineering

FLOWCHART:



CODE:

```
package Assignment8;

import java.io.*;
import java.util.Scanner;

public class FileHandling
{
    public static void main(String[] args)
    {
        Scanner sc = new Scanner(System.in);
        String file = "data.txt";
        try
        {
            System.out.println("Enter text (type 'exit' to stop):");

            FileWriter fw = new FileWriter(file);
            BufferedWriter bw = new BufferedWriter(fw);

            while (true)
            {
                String s = sc.nextLine();
                if (s.equalsIgnoreCase("exit"))
                {
                    break;
                }
                bw.write(s);
            }
        }
        catch (IOException e)
        {
            e.printStackTrace();
        }
    }
}
```

DES PUNE UNIVERSITY
School of Engineering and Technology
Computer Engineering and Technology
Program: B.Tech. Computer Science and Engineering

```
        bw.newLine();
    }

    bw.close();
    System.out.println("Data written to " + file);

    FileReader fr = new FileReader(file);
    BufferedReader br = new BufferedReader(fr);

    System.out.println("\nReading from file:");
    String line;
    while ((line = br.readLine()) != null)
    {
        System.out.println(line);
    }

    br.close();
}
catch (IOException e)
{
    System.out.println("Error: " + e.getMessage());
}
sc.close();
}
```

OUTPUT:

Enter text (type 'exit' to stop):

hi

this is riddhee

exit

Data written to data.txt

Reading from file:

hi

this is riddhee

CONCLUSION:

The program demonstrates Java file handling by writing user input to a file using `FileWriter` and `BufferedWriter`, and then reading it back using `FileReader` and `BufferedReader`. It highlights how data can be stored permanently and retrieved efficiently for use in applications that require data persistence.